

I want to have a baby

Management of infertility



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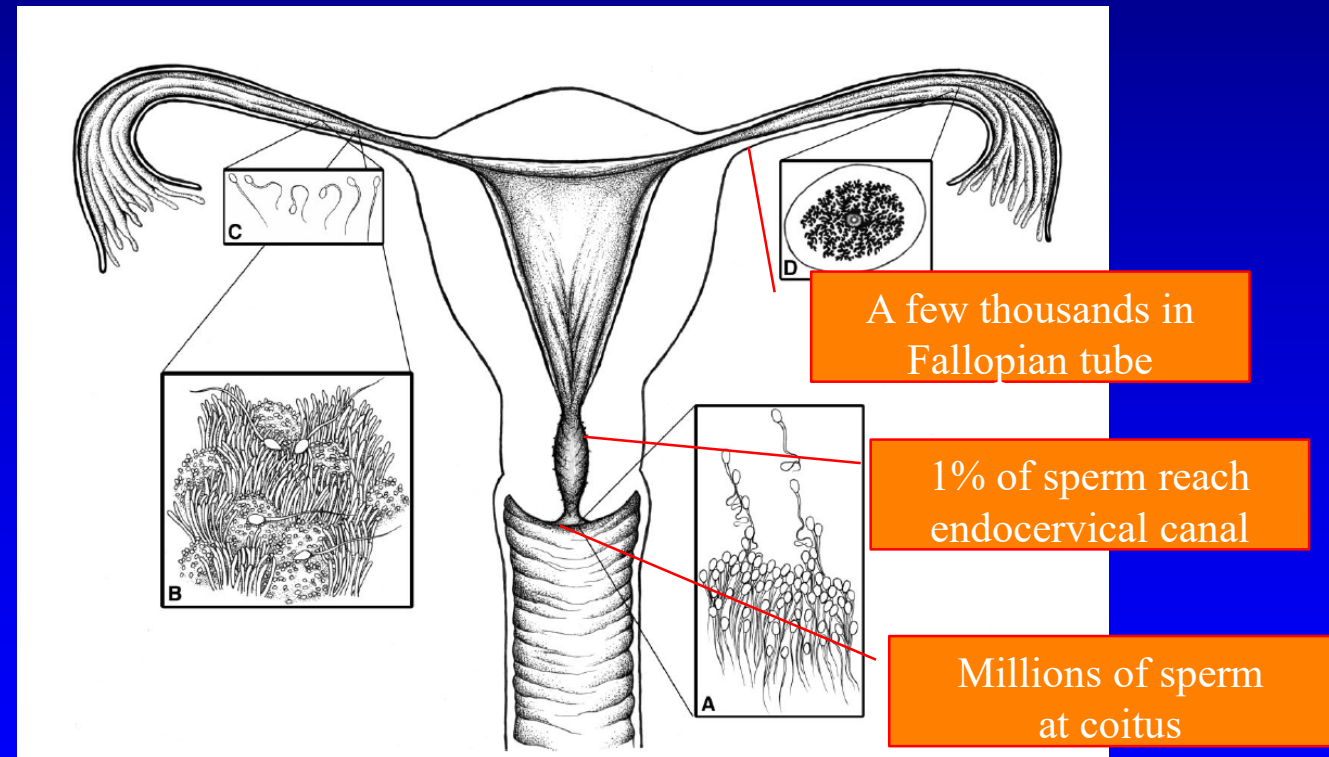
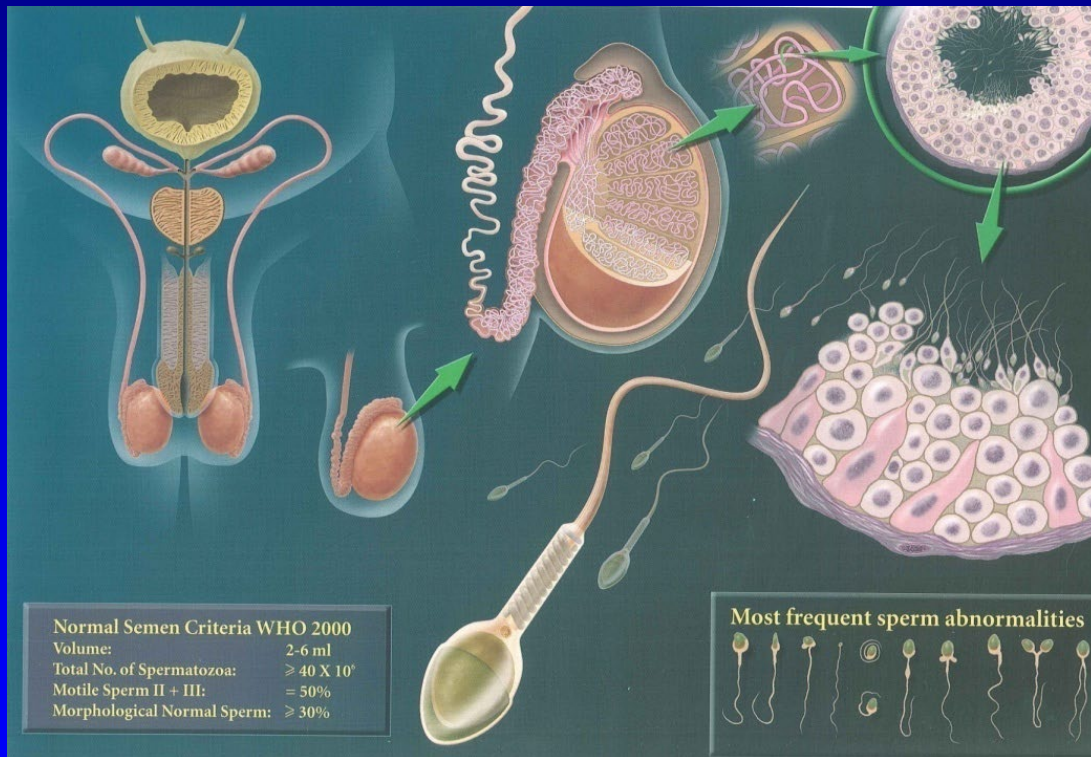
Infertility

1. A disease characterized by failure to establish a clinical pregnancy after 12 months of regular unprotected sexual intercourse
(The International Glossary on Infertility and Fertility Care, 2017)
2. Failure to achieve pregnancy within 12 months of unprotected intercourse in women <35 years or within 6 months in >35 years.
(ACOG and ASRM, 2019)
3. 10-15% of couples affected

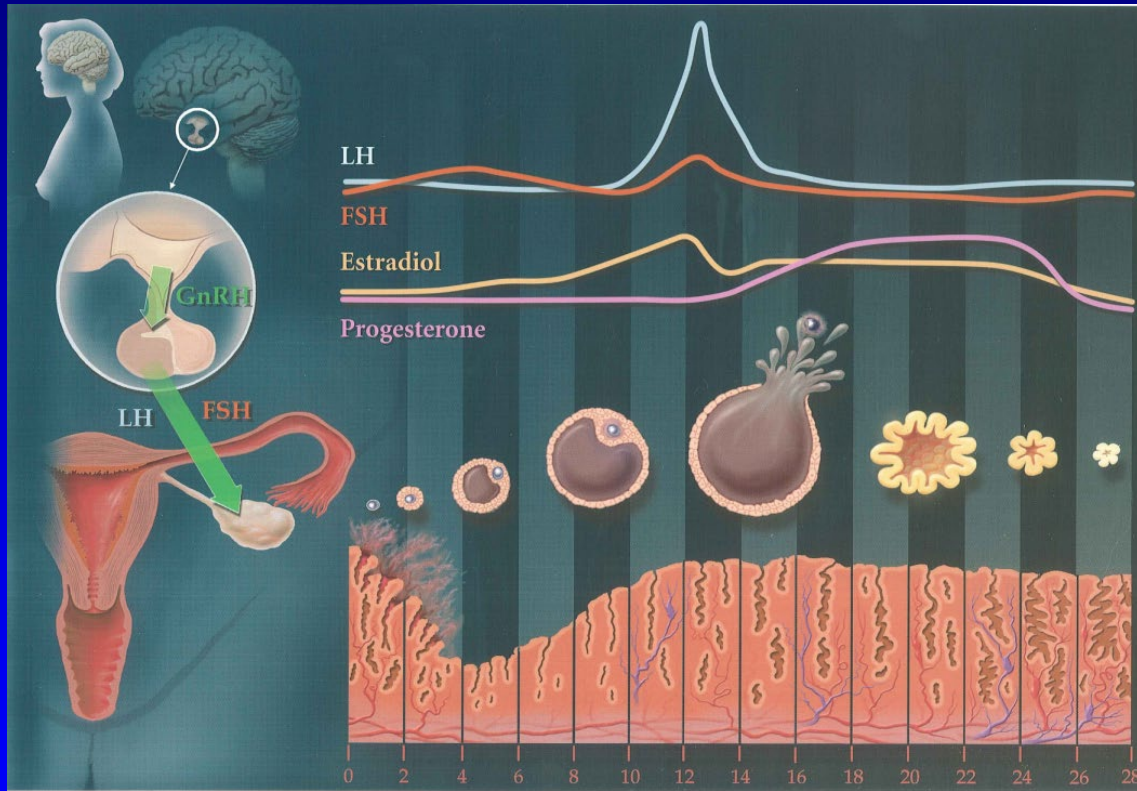
Agenda

1. Normal fertility
2. Causes of infertility
3. Investigations
4. Treatment

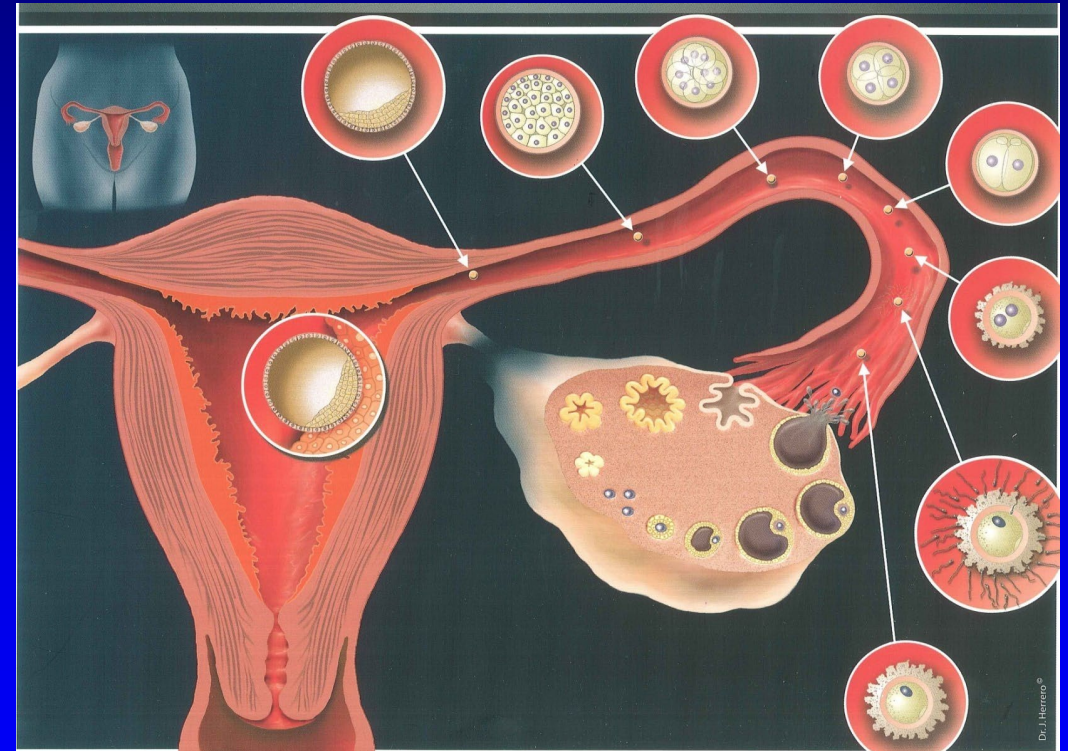
Normal fertility in men: sperm production and journey in upper female genital tract



Normal fertility in women: ovulation, fertilization and embryo implantation



Follicle development and ovulation



Fertilization and implantation

Pregnancy rates

- Peak monthly pregnancy rate ~ 25-30%
 - cumulative pregnancy rate in 1 year ~ 80%
 - cumulative pregnancy rate in 2 years ~ 90%
- Important in the interpretation of pregnancy rate by assisted methods
- Cumulative pregnancy rate

Causes of infertility

- Both women and men can be affected
- Multiple factors are common



Causes of infertility

Female factors



- ❑ ovulatory (15%)
- ❑ tubal (20%)
- ❑ endometriosis (25%)
- ❑ others: cervical, immunological, coital

Male factors (30%)



- ❑ abnormal semen due to production defects e.g. idiopathic, endocrine, trauma, genetic
- ❑ no sperm due to obstructive defects e.g. absent vas, vasectomy
- ❑ coital

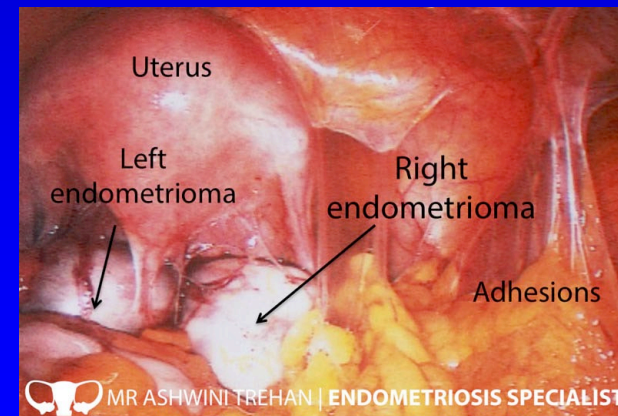
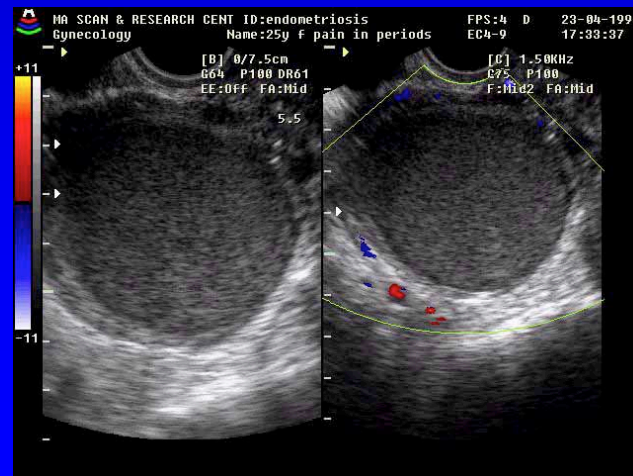
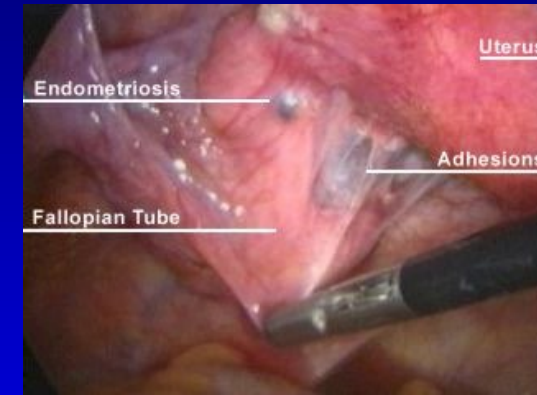
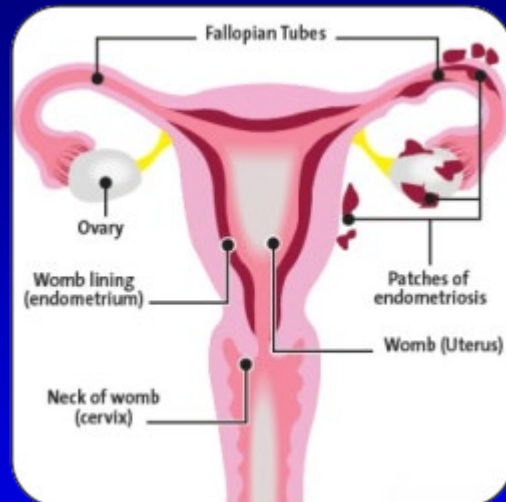
Unexplained

- By exclusion in the presence of ovulation, patent tubes and normal semen

Five important causes

1. Ovulatory dysfunction/anovulation
2. Tubal problems
3. Endometriosis
4. Male factors
5. Unexplained

Endometriosis



History

Female ♀

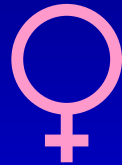
- Age
- Menstrual cycle
 - Regularity (21-35 days)
- History of pelvic infection or surgery
- Previous investigations and treatment

Male ♂

- Age / occupation
- Past health
- Coital history
- Smoking/alcoholic

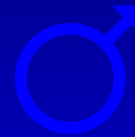
Physical examination

Female



- Body weight
- Vaginal examination
 - uterine size
 - mobility
 - adnexal mass

Male



- ? Necessary
- Testicular size
- Vas and epididymis
- Varicocele

Early referral

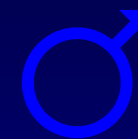
Female

- >35 years old
- Irregular cycles
- Previous pelvic surgery
- Previous sexual transmitted disease (STD)
- Abnormal pelvic examination

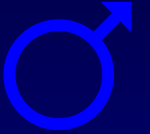
Male

- Systemic illness
- Previous genital pathology
- Previous STD
- Varicocele
- Abnormal genital examination

Investigations– Semen analysis

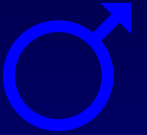


Semen analysis

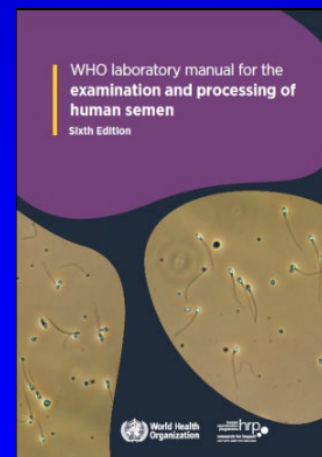
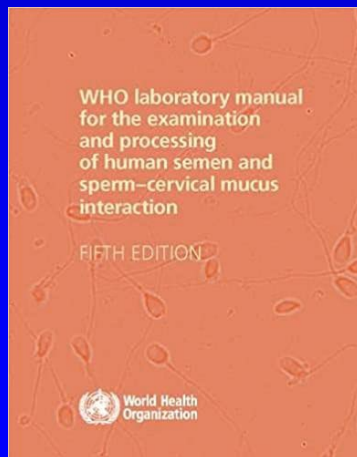


- 2 to 3 samples after 2-7 days of sexual abstinence
- WHO criteria
- Low predictive values

WHO manual 5th and 6th Edition



- Reference group: fertile men whose partners who have regular menstrual cycles and bilateral patent tubes got pregnant within 12 months
- 5th centiles as the lower reference values



WHO reference group and values

Origin of study	Number of subjects	Reference to publication containing data	Data obtained
New subjects since WHO 2010	1789		
Italy, Europe	105	Lotti et al. (2020) ⁴	Yes
Iran, Asia ^a	168	Aboutorabi et al. (2018) ⁵	Yes
Egypt, Africa	240	Zedan et al. (2017) ⁶	Yes
China, Asia	1200	Tang et al. (2015) ⁷	Yes
Greece, Europe	76	Evgeni et al. (2015) ⁸	Yes
Subjects used in WHO 2010	1800		
Australia, Oceania	206	Stewart et al. (2009) ¹⁹	Yes ^b
Norway, Europe	82	Haugen et al. (2006) ²⁰	Yes ^b
United States of America, Americas	487	Swan et al. (2003) ²¹	Yes ^b
France, Finland, Denmark, United Kingdom, Europe	826	Auger et al. (2001) ²² , Jørgensen et al. (2001) ²³ , Slama et al. (2002) ²⁴	Yes ^b
Denmark, Europe	199	Bonde et al. (1998) ²⁵ , Jensen et al. (2001) ²⁶	Yes ^b

	5 th Edition	6 th Edition
Semen volume (ml)	1.5	1.4
Concentration (x10 ⁶ /ml)	15	16
Total sperm no. (x10 ⁶)	39	39
Total motility (%)	40	42
Progressive motility (%)	32	30
Morphology (%)	4	4

Campbell MJ et al. Distribution of semen examination results 2020 - A follow up of data collated for the WHO semen analysis manual 2010. Andrology. 2021 May;9(3):817-822.

Sperm concentration in natural conception

Table 2 Exposure and Pregnancies According to Different Sperm Concentrations While Couples Used T Enanthate Alone for Contraception

Sperm concentration stratum	Concentration	Exposure	Pregnancies	Rate per 100 person-years*
	$\times 10^6 / \text{mL}$	<i>person-years</i>		
Initial threshold for efficacy ----->	>5.0	1.1	2	189 (22.9 to 683)
	4.1 to 5.0	1.4	4	291 (79.3 to 745)
Revised threshold for efficacy ----->	3.1 to 4.0	2.3	1	42.7 (1.1 to 238)
	2.1 to 3.0	3.5	1	28.3 (0.7 to 158)
	1.1 to 2.0	6.9	1	14.5 (0.4 to 81.0)
	0.1 to 1.0	39.1	2	5.1 (0.6 to 18.5)
Azoospermia	0.0	230.4	0	0.0 (0.0 to 1.6)
Severe oligozoospermia	0.1 to 3.0	49.5	4	8.1 (2.2 to 20.7)
Azoospermia and severe oligozoospermia	0.0 to 3.0	279.9	4	1.4 (0.4 to 3.7)

* Values in parentheses are 95% CI.

- 399 fertile men with sperm concentration >20 M/ml
- Sperm concentration after male hormonal concentration
- Threshold for efficacy: 3 M/ml

WHO Task Force on Methods for the Regulation of Male Fertility. Contraceptive efficacy of testosterone-induced azoospermia and oligozoospermia in normal men. *Fertil Steril*. 1996 Apr;65(4):821-9.

Semen analysis in fertile and infertile men

- 696 fertile couples Vs 765 infertile couples
 1. Infertile couples: unable to conceive for at least 12 months
 2. Fertile couples: pregnant or deliver a baby within 2 years
- Women had ovulation and patent tubes confirmed

Semen analysis in fertile and infertile men

Mean+/-SD	Fertile group	Infertile group
Concentration ($\times 10^6/\text{ml}$)	67 ± 50	52 ± 42
Motility (%)	54 ± 13	49 ± 15
Morphology (%)	14 ± 5	11 ± 6

- Extensive overlap between fertile and infertile
- None of parameters diagnostic of infertility

Causes of male infertility

1. Disorders of testicular control

- Hypothalamic-pituitary disorders
- Hypogonadotropic hypogonadism
- Hyperprolactinaemia

2. Obstructive disorders

- Infections
- Congenital absence of vas
- Surgery - vasectomy

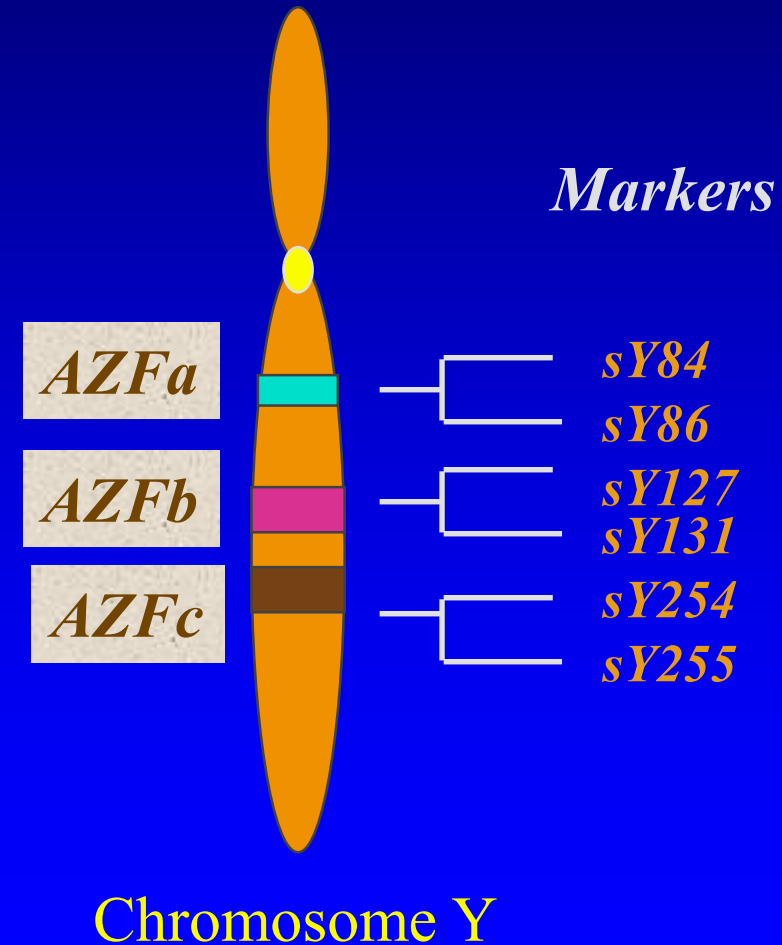
3. Primary testicular disorders

- Chromosomal: Klinefelter's Syndrome (47XXY)
- Varicocele or testicular hyperthermia
- Infections: mumps
- Trauma
- Cryptorchidism
- Previous radiotherapy/chemotherapy
- Chronic illness

Genetic causes of male infertility

Y chromosome microdeletions

- 3 deletion regions (AZFa-c) of Yq11 linked with male infertility
- Different deletion regions affect distinct and separate phases of spermatogenesis
- No sperms found in AZFa and AZFb



Investigations of male infertility

1. Hormonal assays: FSH, prolactin, testosterone
2. Karyotype and Y microdeletion for testicular failure
3. Vasogram
4. Testicular biopsy: not as a routine; may be of prognostic value

Investigations--Ovulation



- Serum mid-luteal progesterone levels (a week before next expected period)
- Irregular cycles
 - ☐ FSH, prolactin, thyroxine
 - ☐ Ultrasound--ovarian morphology (PCO)
- Regular cycles
 - ☐ prolactin or thyroxine not indicated



Tests of ovulation

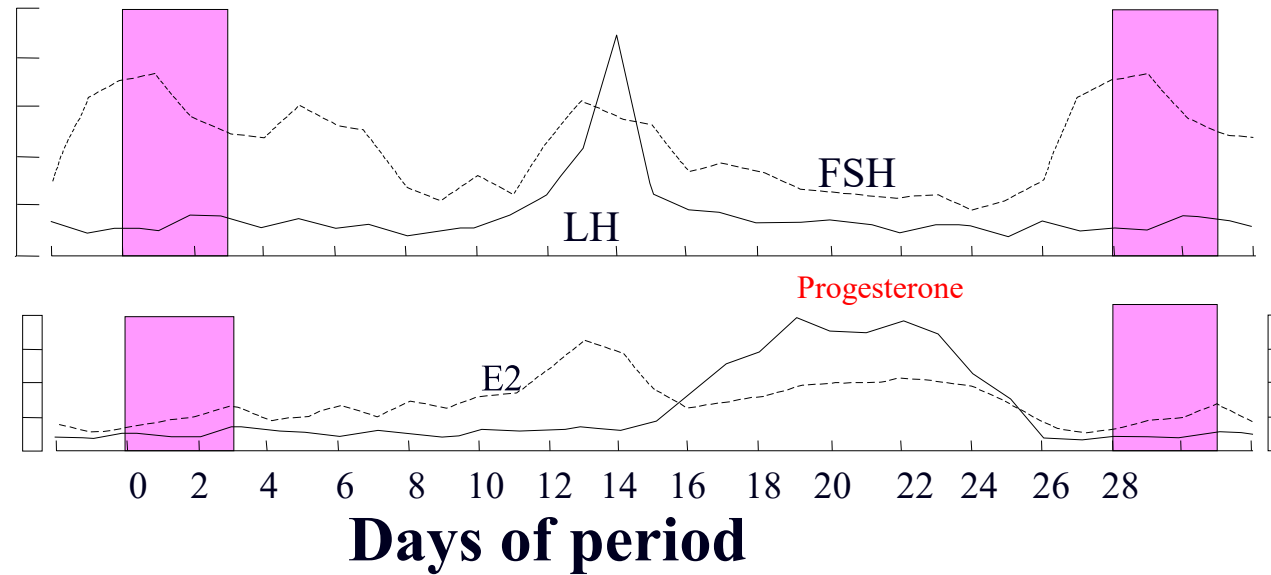
- Gold standard is pregnancy
- Other tests inferential
- Regular cycles (21-35 days) suggest 95% chance of ovulation
(Behre et al, HR 2000)
- Serum progesterone level, BBT, urine LH kits, pelvic ultrasound, endometrial biopsy



Serum progesterone level

- Taken in the mid-luteal phase, i.e. ~1 week before the expected onset of the next period
- >3 ng/mL or >10 nmol/L as presumptive and sufficient evidence of recent ovulation (Wathen 1984).
- Serum P levels can fluctuate by sevenfold over a few hours
- Not to assess quality of luteal phase

Basal body temperature

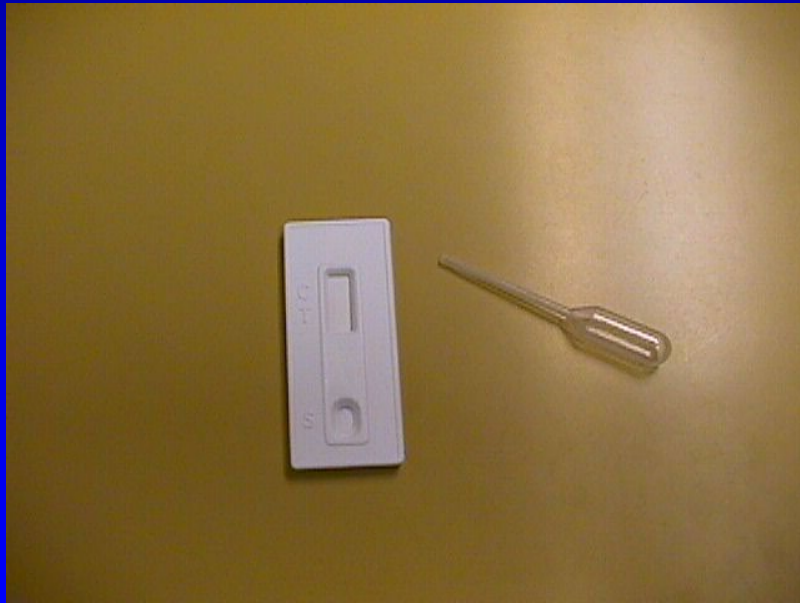


Basal body temperature

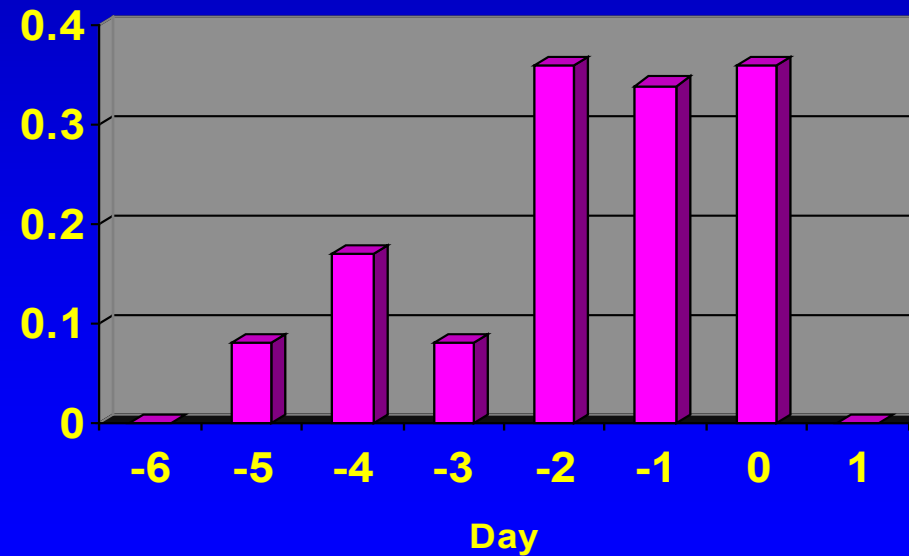


- Around similar time over 3-4 months
- BBT nadir as evidence of ovulation
- Problem:
 - Not accurate because ovulation was between 6 days before and 4 days after nadir.
 - Difficult to interpret in many cases
 - Time the intercourse

Urinary LH kits



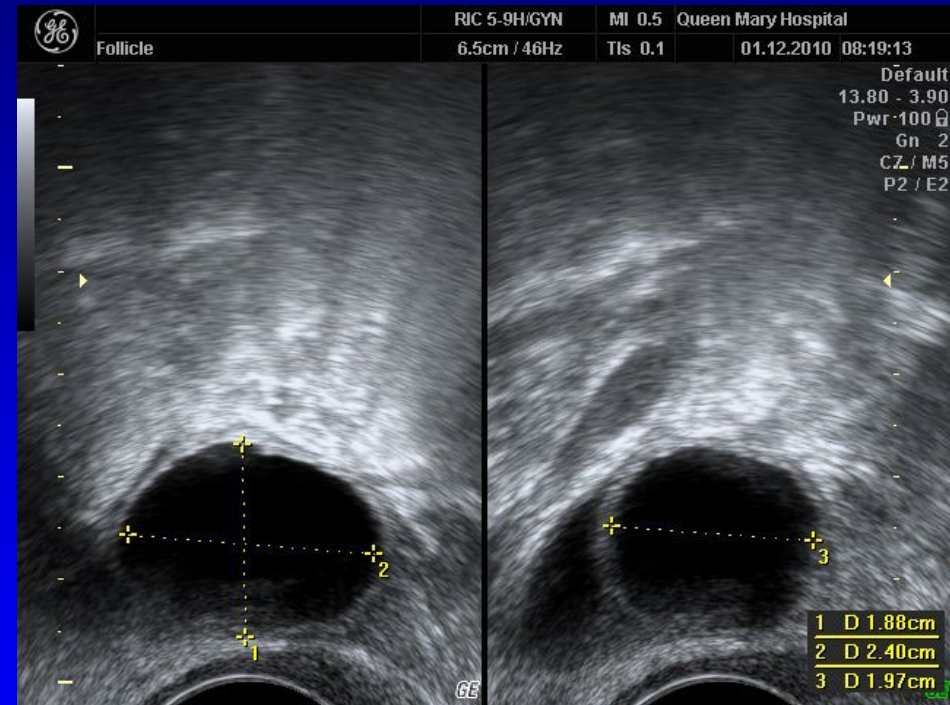
Probability of conception on specific days near the ovulation day



Pelvic ultrasound



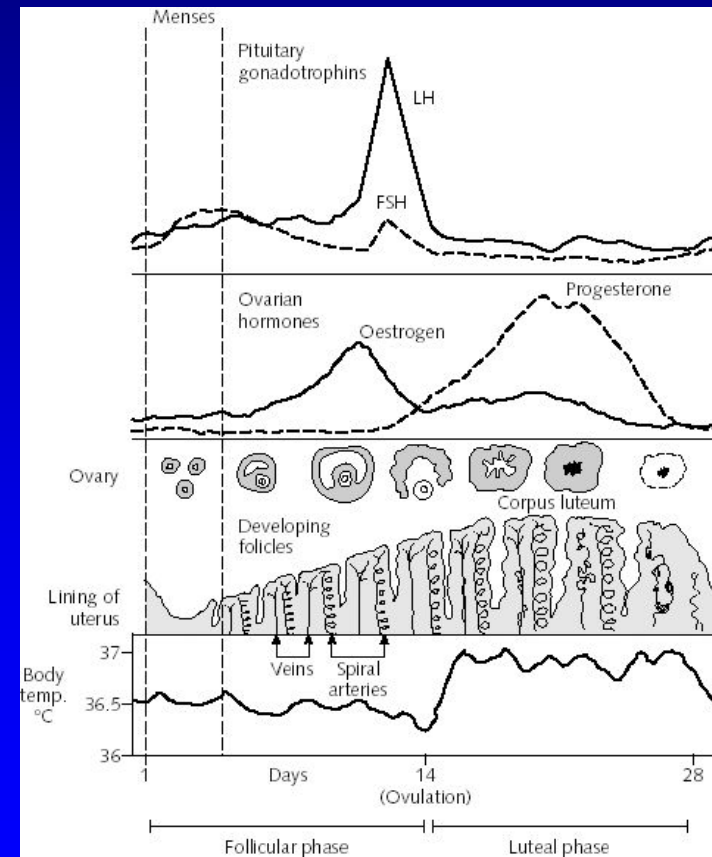
- Progressive follicular growth, sudden collapse of the pre-ovulatory follicle, a loss of clearly defined follicular margins, the appearance of internal echoes within the corpus luteum, and an increase in cul-de-sac fluid volume (de Crespigny 1981)
- Repeated scanning; labour intensive



Ovarian cycle



Hypothalamic-pituitary-ovarian axis



Causes of anovulation



1. Hypothalamus

- Weight change: obesity, anorexia nervosa
- Drugs: tranquillizers
- Psychological disturbance

2. Pituitary

- Tumours
- Sheehan's syndrome
- Hyperprolactinaemia

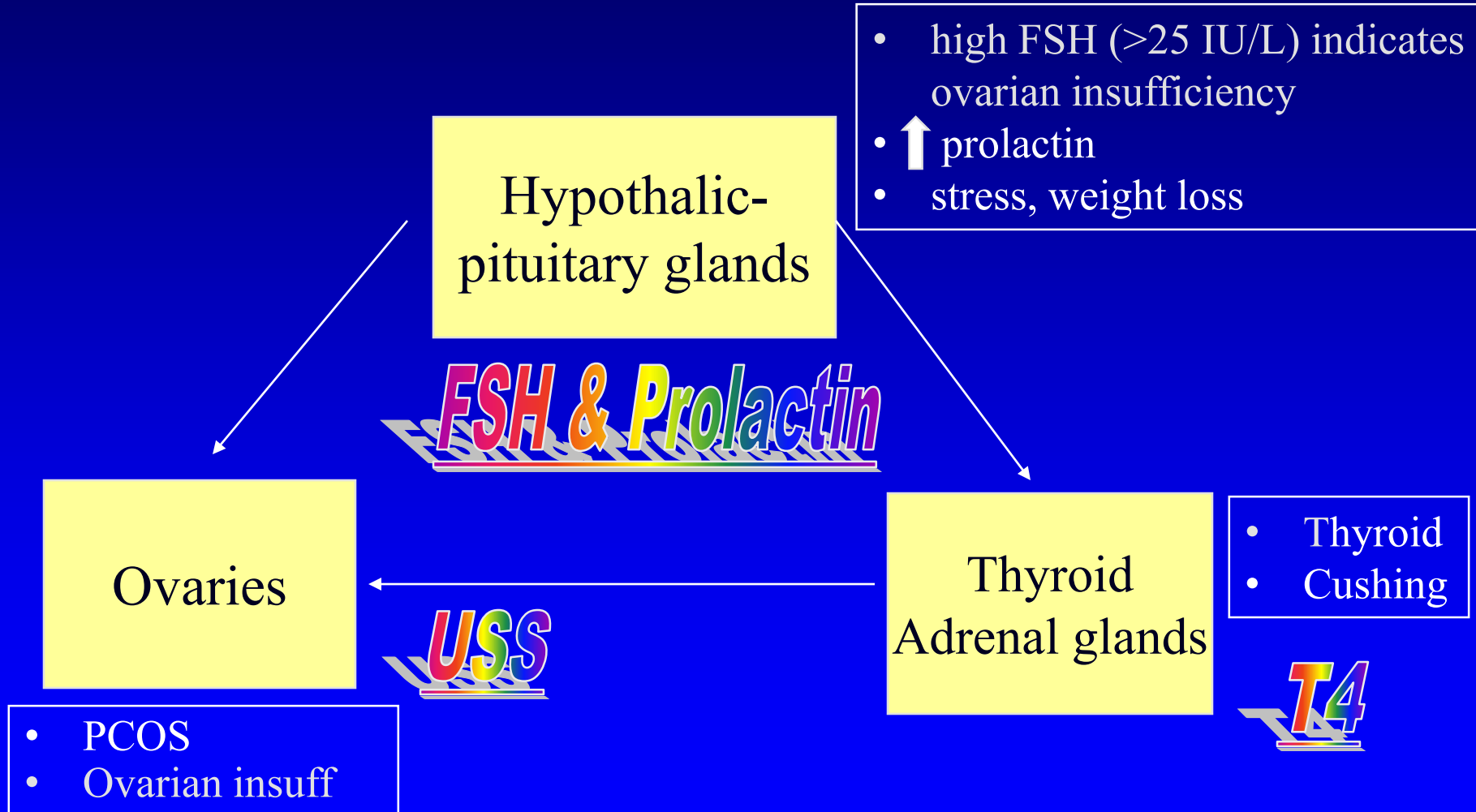
3. Ovary

- ovarian insufficiency: chromosomal disorders (Turner Syndrome), surgery, radiotherapy, chemotherapy, mumps
- polycystic ovary syndrome (PCOS)

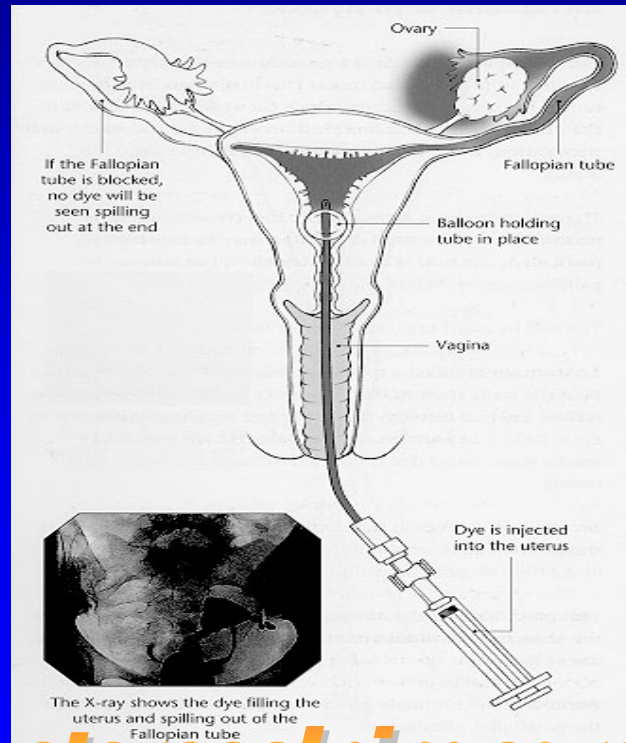
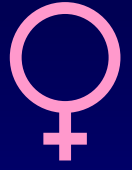
4. Other endocrine disorders

- thyroid disorders
- adrenal disorders

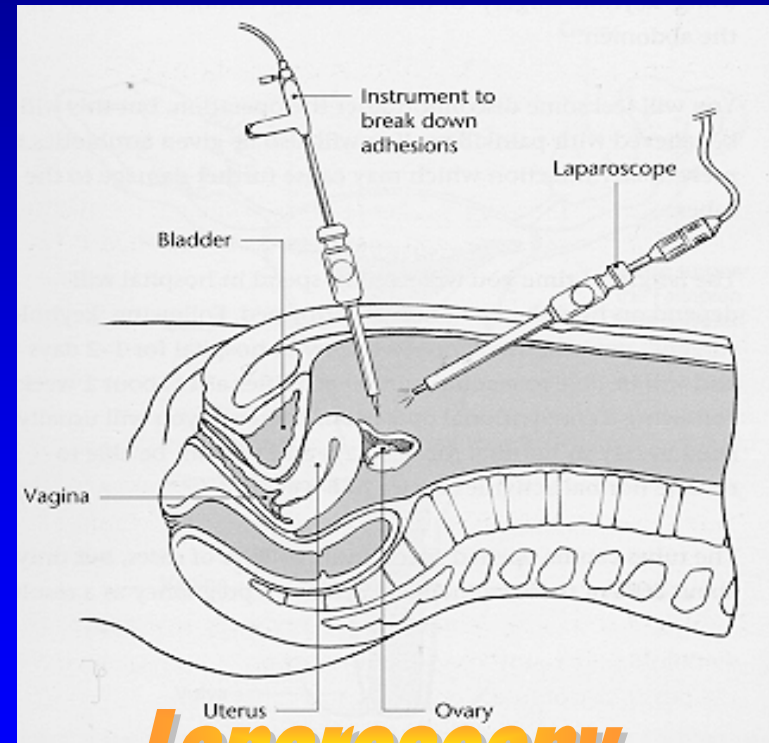
Investigations of anovulation



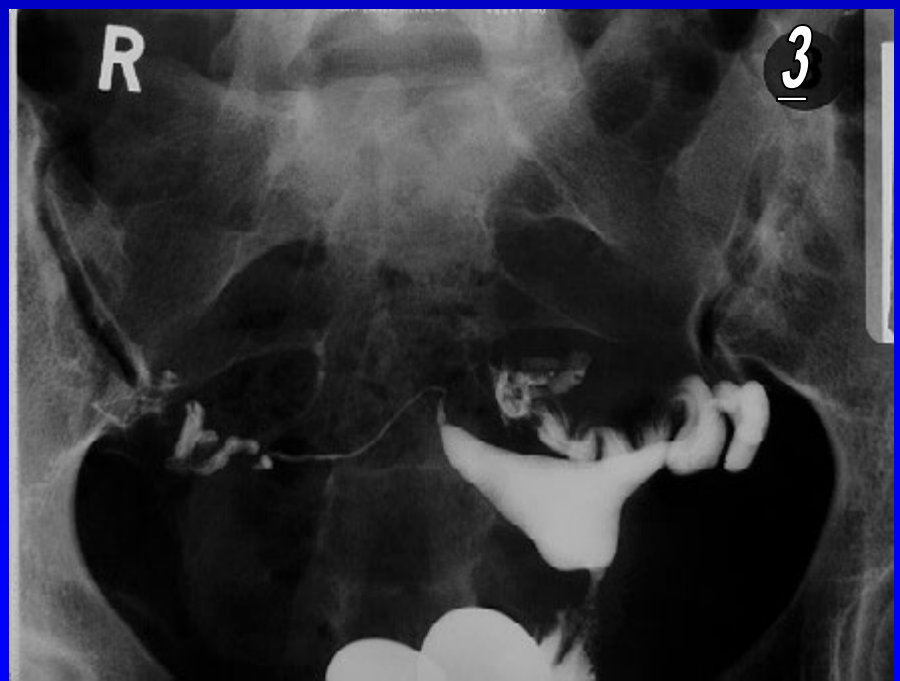
Investigations—Tubal patency test



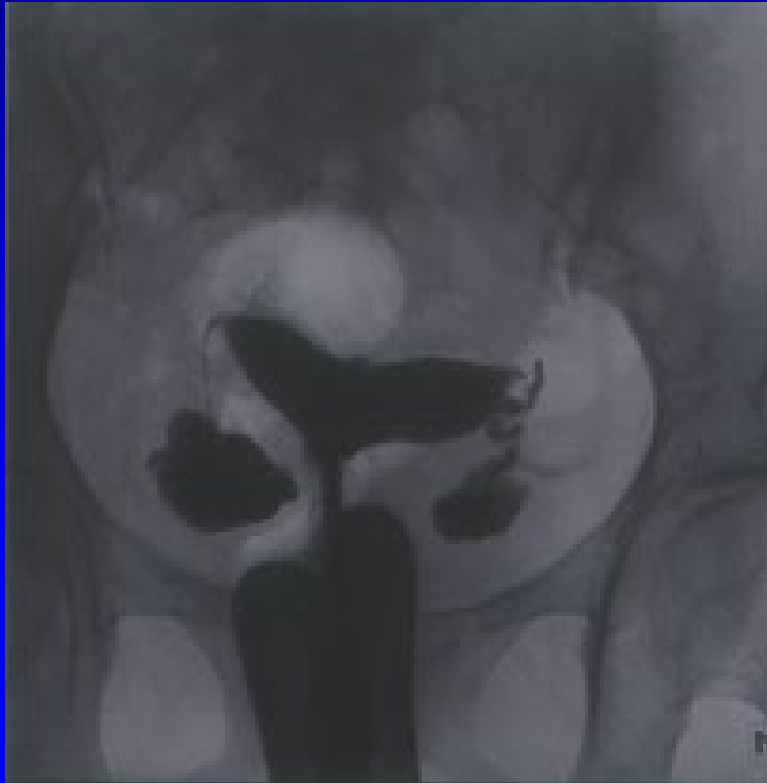
Hysterosalpingogram



Laparoscopy



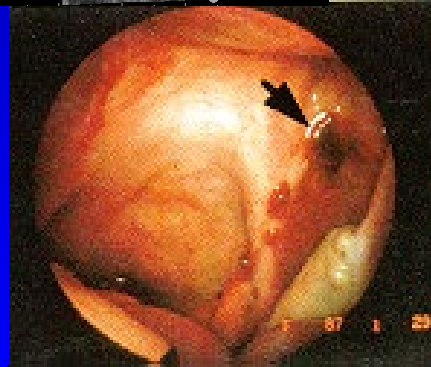
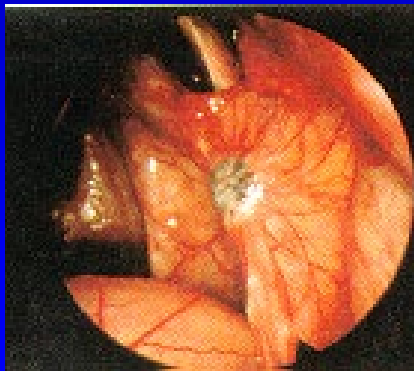
Hydrosalpinx



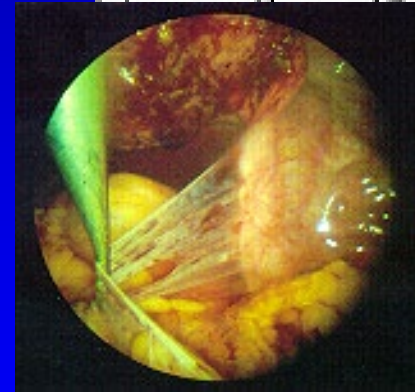
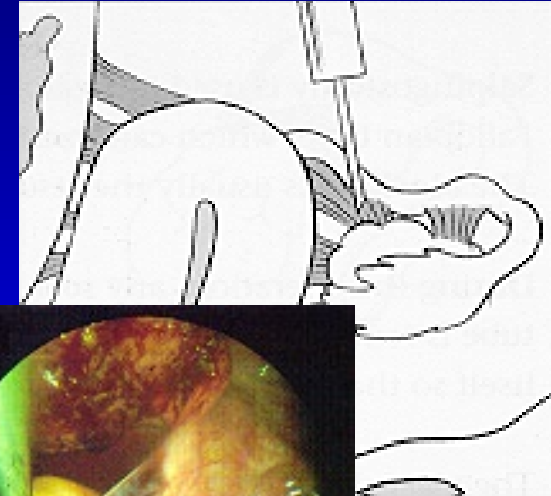
Hysterosalpingogram

- False +ve because of spasm in proximal ends
- Peritubal adhesion not detected
- 4-5% pelvic inflammatory disease after hysterosalpingogram

Laparoscopy-diagnostic / therapeutic



Endometriosis



Pelvic adhesion

Tubal patency test

- Hysterosalpingogram (HSG)

- Less invasive and outpatient procedure
- Can assess uterine cavity

- Laparoscopy

- Operative procedure requiring general anaesthesia
- More accurate
- Assess & treat endometriosis/adhesions

- HSG: women with no comorbidities such as previous PID or ectopic pregnancy; clinical s/s of endometriosis
- Hysterosalpingo-contrast-ultrasonography may be an alternative to HSG
- Laparoscopy: women thought to have comorbidities

Investigations that should NOT be routinely ordered

- Laparoscopy for unexplained infertility
- Advanced sperm function testing (eg, DNA fragmentation testing)
- Postcoital testing
- Thrombophilia testing
- Immunologic testing
- Karyotype
- Endometrial biopsy
- Prolactin

ACOG, 2019

Pre-pregnancy advice



- 0.4 mg **folic acid** whilst trying to conceive and during the first 12 wks of pregnancy to prevent neural tube defects
- Reduce body weight in obese women
- Stop smoking
- Stop alcohol

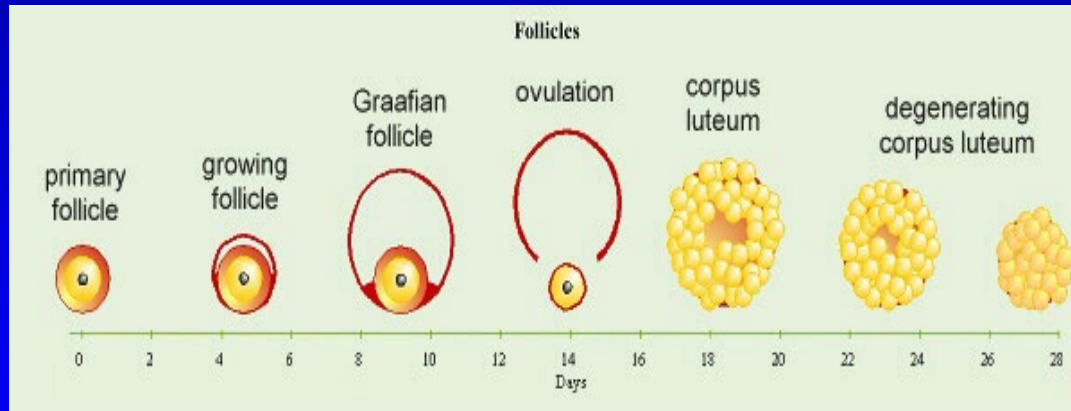
Pre-pregnancy advice (Male)

- Stop smoking
- Avoid excessive alcohol
- Men with poor quality sperm advised to
 - wear loose fitting underwear and trousers and
 - avoid occupational or social situations that might cause testicular hyperthermia

Ovulation induction

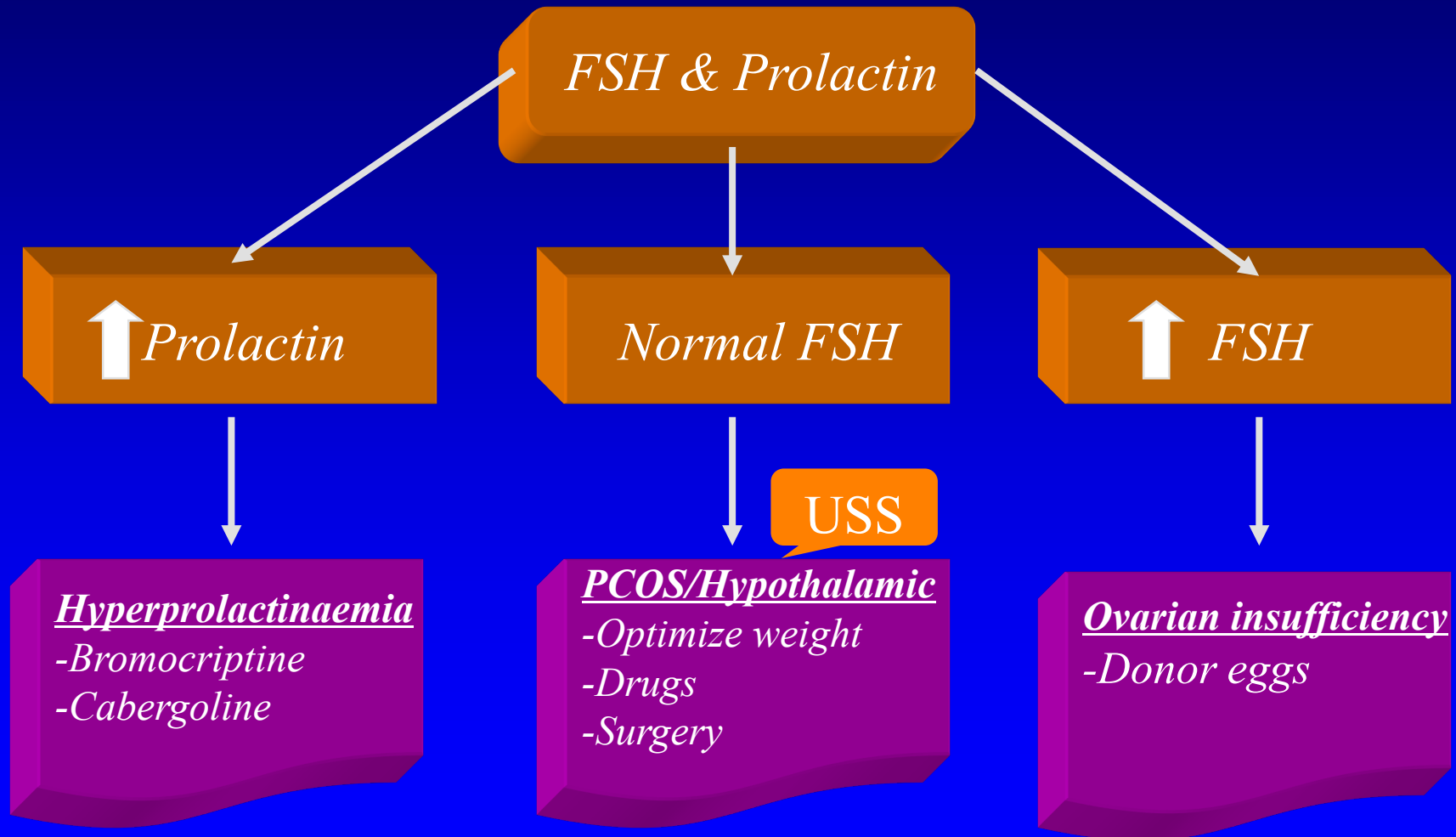


- Aim: development of a single follicle



- Weight reduction
- Drugs
 - Letrozole / clomid
 - Gonadotrophin
 - Bromocriptine
 - Others: insulin sensitising agents (metformin)
- Surgery
 - ovarian drilling

Ovulation Induction



Letrozole

- First line oral therapy in PCOS
- Aromatase inhibitors
- 2.5-7.5 mg/ day for 5 days
- Ovulation and pregnancy rates higher than clomid
- Complications/side effects: less multiple pregnancy when compared with clomid

Clomiphene citrate (CC)

- Anti-oestrogen acting at hypothalamus
- 50-150 mg/ day for 5 days; up to 6 cycles
- Complications/side effects: hot flushes, multiple pregnancy, ovarian cysts, abdominal distension/pain, blurring of vision
- Ovulation rate: 50-80%
- Pregnancy rate 30-50%

Gonadotrophin

- FSH & LH or recombinant FSH injections
- Acts directly on ovaries and very effective
- Risk of OHSS and multiple pregnancy high
- Requires careful monitoring

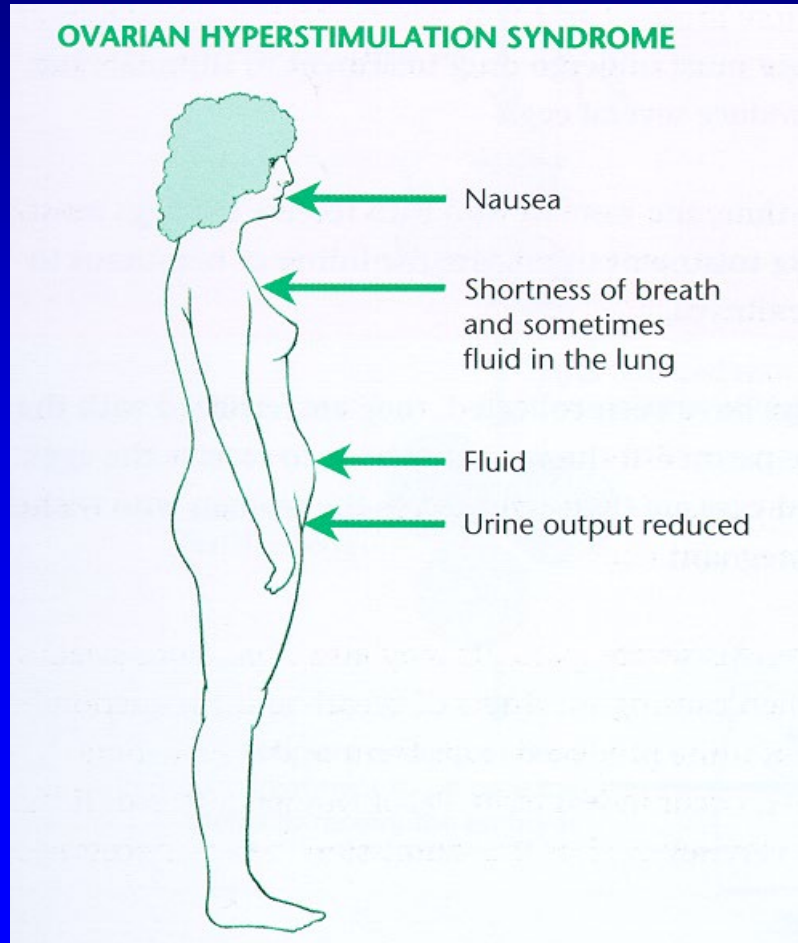
Multiple pregnancies



Up to 40-50% !!



Ovarian hyperstimulation Syndrome



- 1-5 %
- life-threatening condition

Bromocriptine

- Dopamine agonist useful in patients with hyperprolactinaemia
- Ovulation rate over 90%
- Pregnancy rate 80%
- Side effects: nausea, postural hypotension

Ovarian Insufficiency

- Medical treatment is not effective
- Oocyte donation
 - Oocyte obtained from a donor
 - Oocyte fertilized by the husband's sperms
 - Embryo replaced into the recipient's uterus

Tubal factors

- Tubal surgery

- microsurgical technique
- laparotomy or laparoscopy
- adhesiolysis, re-anastomosis, salpingostomy
- results

- In vitro fertilization (IVF)

Tubal surgery Vs IVF

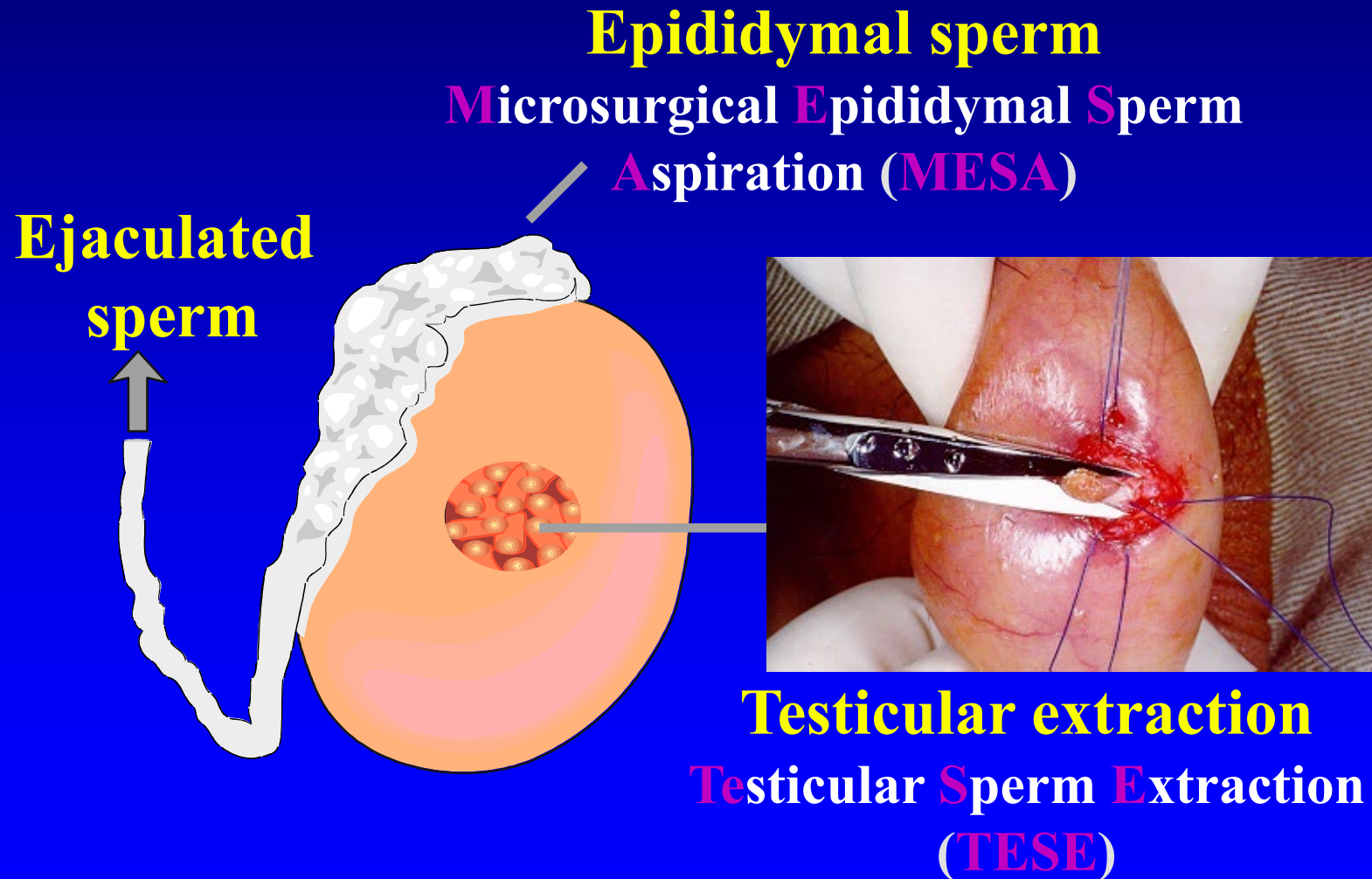
1. The lesions
2. The infertile couple
3. The medical service

Male infertility

Effective medical treatments

- Varicocele treatment
- Vasectomy reversal/overcome correctable obstruction
- Gonadotrophins or GnRH for hypogonadotrophic hypogonadism
- Bromocriptine for sexual dysfunction associated with hyperprolactinaemia
- Ovarian stimulation and intrauterine insemination (IUI)
- IVF with conventional insemination or intracytoplasmic sperm injection (ICSI)

Sources of sperm for ICSI



Male infertility

Ineffective Rx or Rx of Doubtful Value

- Anti-oestrogens, androgens, bromocriptine & kinin-enhancing drugs for abnormalities of semen quality
- Antioxidants, mast cell blockers & alpha blockers need further evaluation
- Systemic corticosteroids for antisperm antibodies

Minimal / mild endometriosis

1. Medical treatment does not enhance fertility
2. Surgical ablation (diathermy or laser) improves fertility in infertile women
3. Ovarian stimulation and IUI
4. IVF

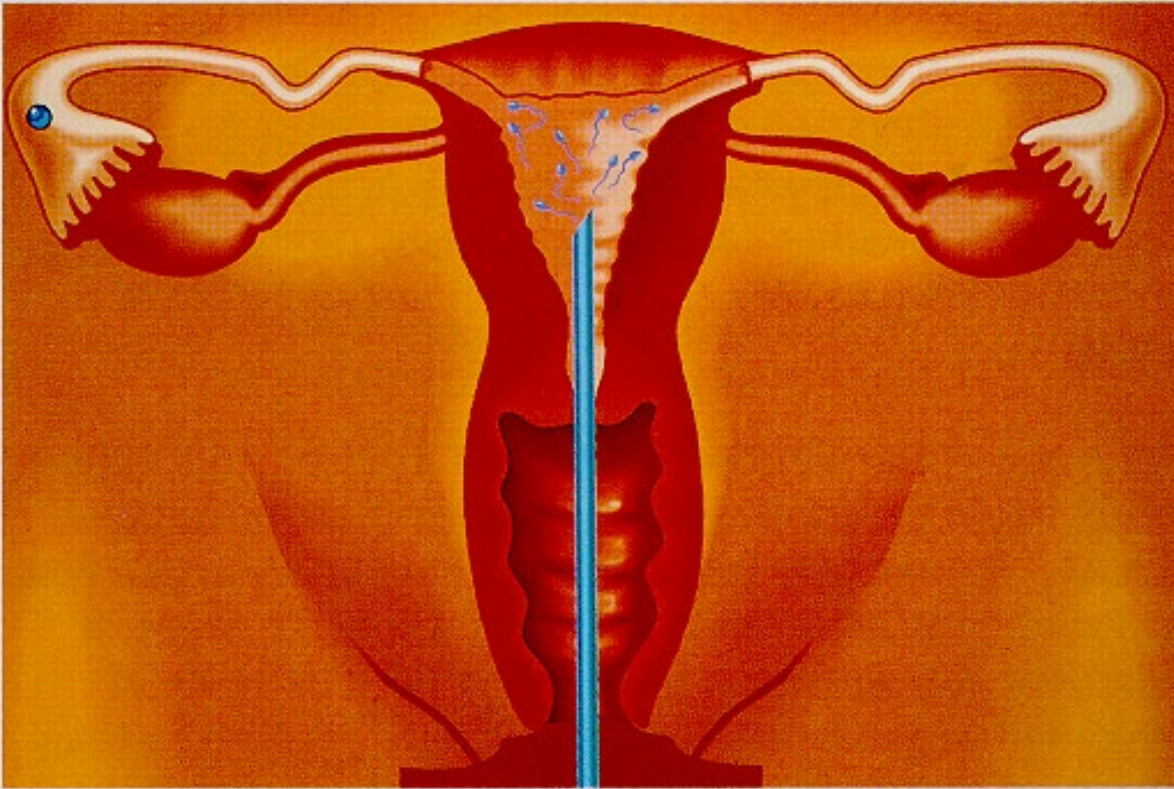
Unexplained infertility

1. Expectant treatment
2. Clomid NOT recommended
3. Unstimulated IUI NOT recommended
 - CC and unstimulated IUI unlikely to offer superior live birth rates compared with expectant management in unexplained infertility, . (Bhattacharya et al., 2008)
4. IVF

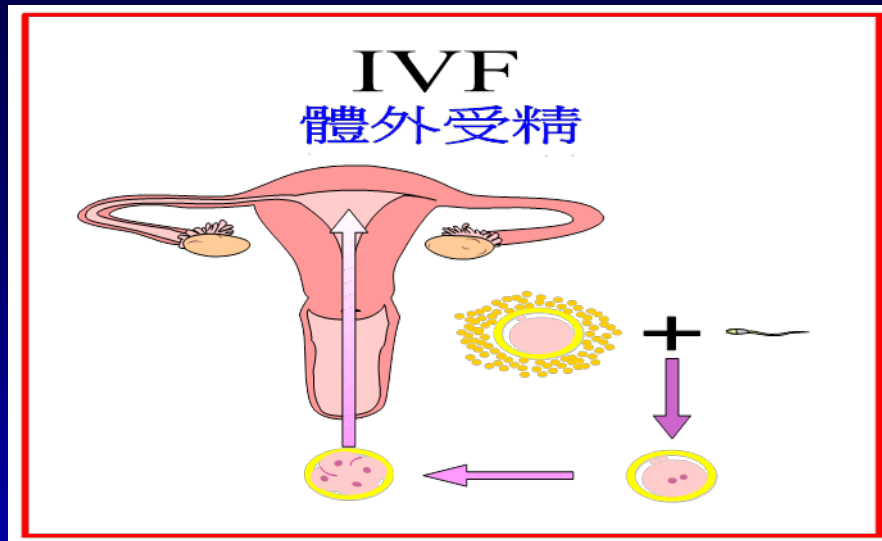
Treatment of coital problem

1. Investigate cause and treat accordingly e.g.
 - psychotherapy
 - Drugs for erectile dysfunction e.g. viagra
2. IUI by husband's sperms at the time of ovulation

Intrauterine insemination (IUI)

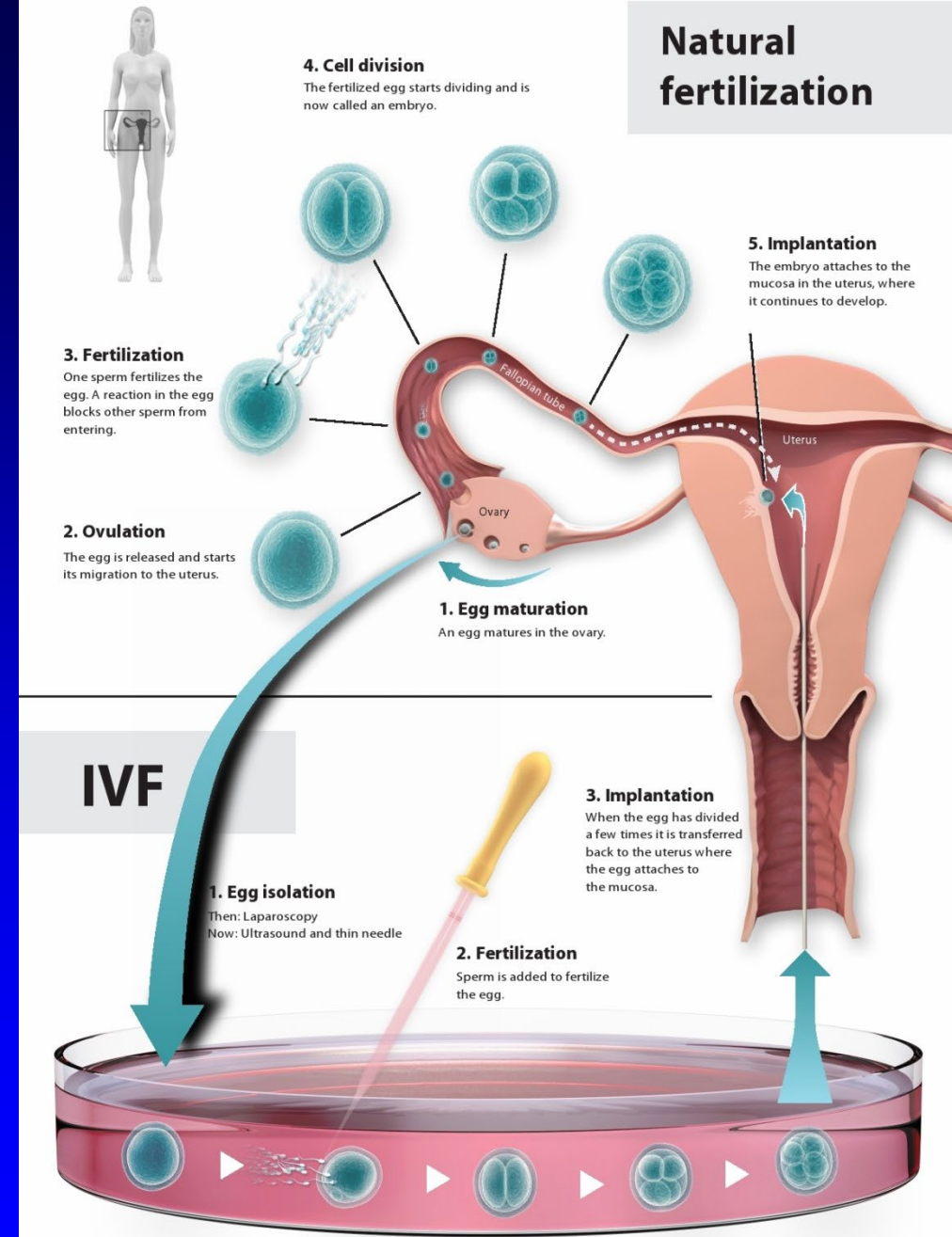


1. Increasing the number of motile sperm in the uterus through IUI around ovulation
2. Ovarian stimulation: increase the number of oocytes available for fertilization
3. Pregnancy rate: ~10% per cycle



IVF procedures

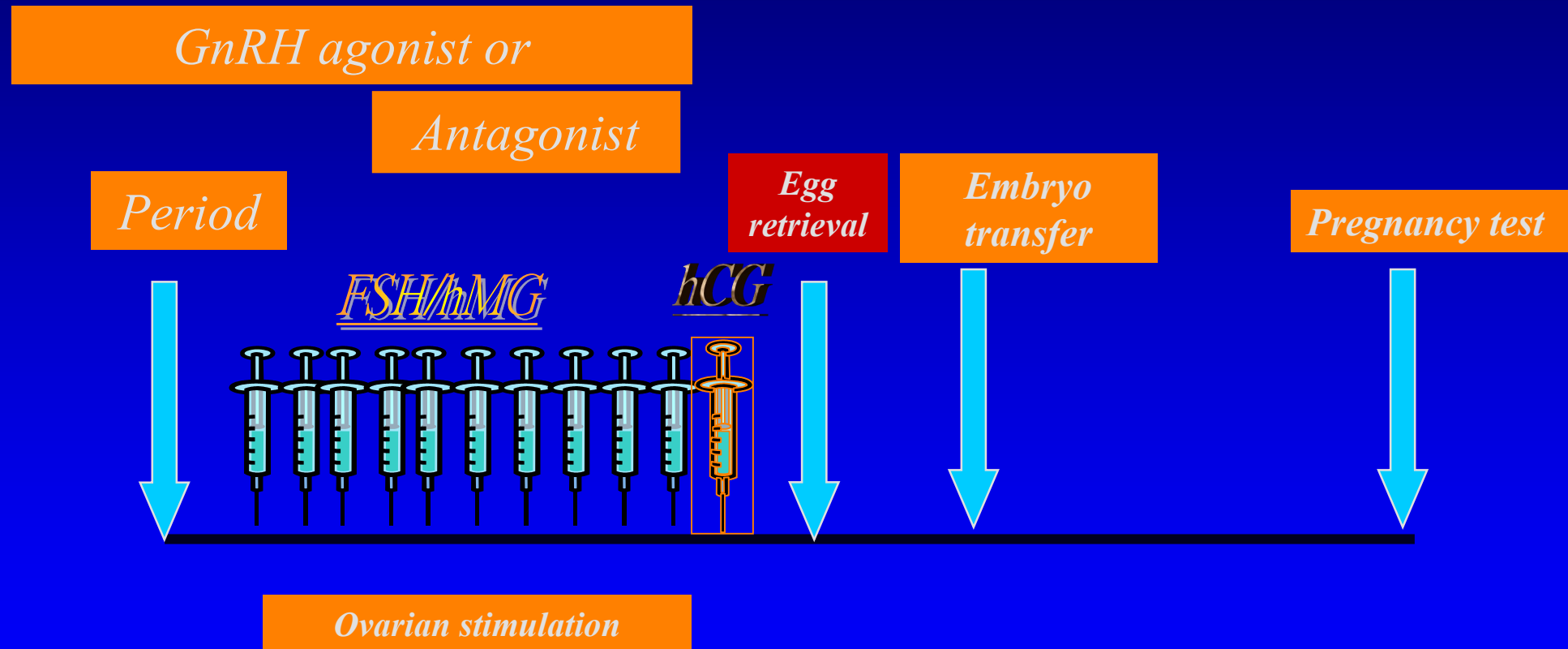
1. Stimulation of ovaries
2. Oocyte pickup
3. Fertilization
4. Embryo transfer



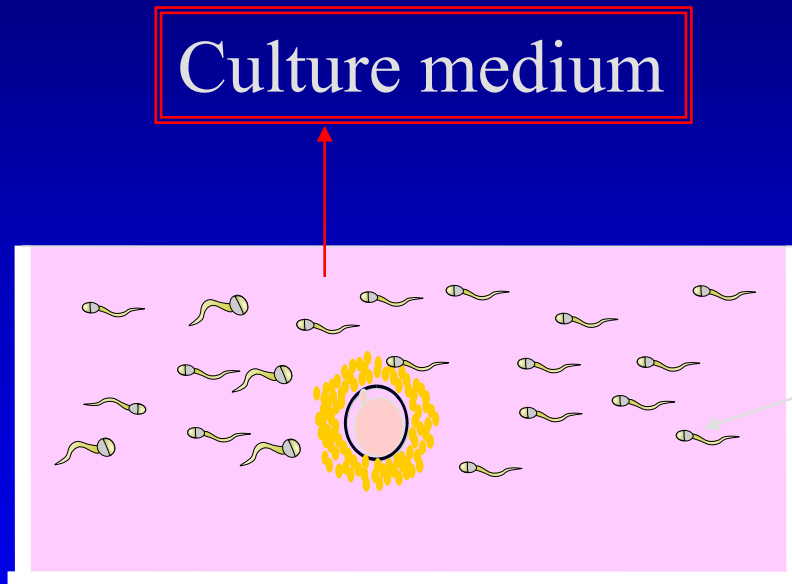
IVF is used when sperm and egg can not meet under normal conditions. Common causes include obstructed fallopian tubes, too few eggs or impaired production of sperm.

© The Nobel Committee for Physiology or Medicine 2010
Illustration: Mattias Karlén

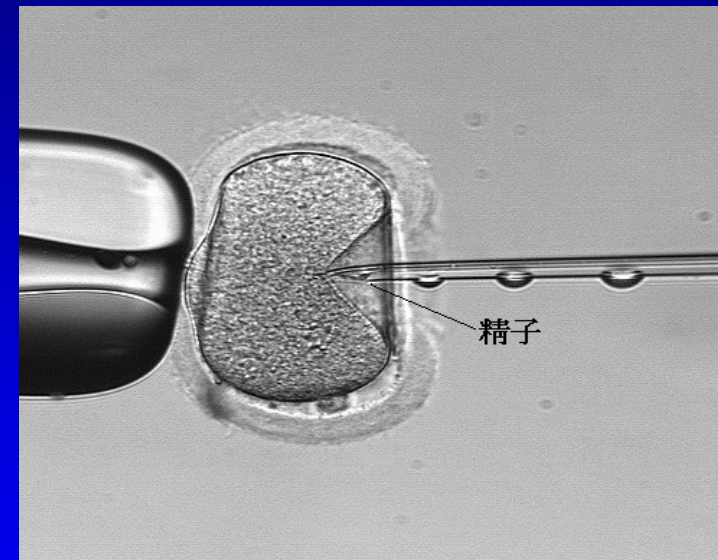
Ovarian stimulation



Two fertilization methods



1. Conventional insemination



2. Intracytoplasmic sperm injection-
injecting single sperm into an egg

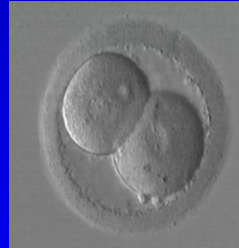
Embryo development



Day 0



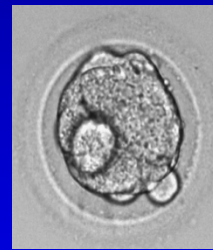
Day 1



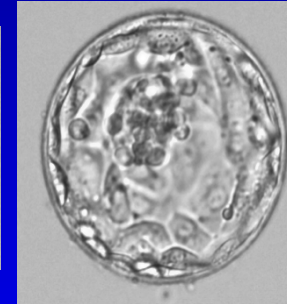
Day 2



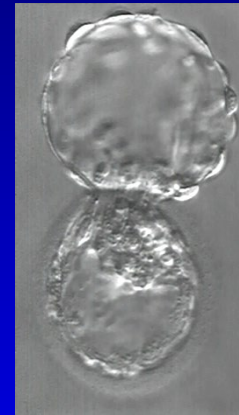
Day 3



Day 4

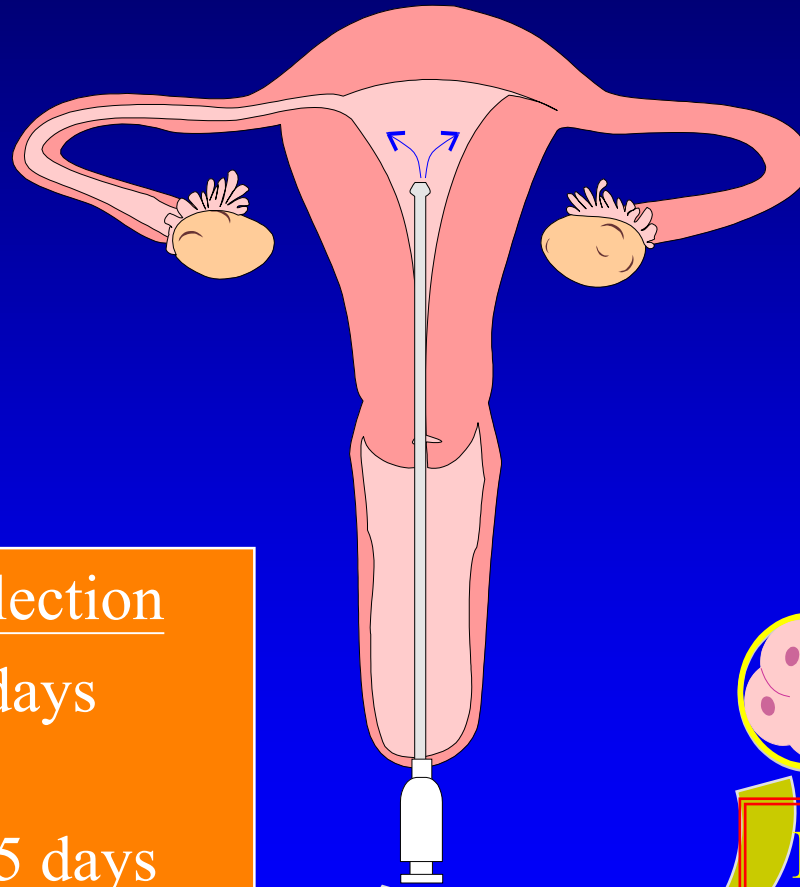


Day 5



Day 6

Embryo transfer



After egg collection

- Embryo (2 days later) or
- Blastocyst (5 days later)

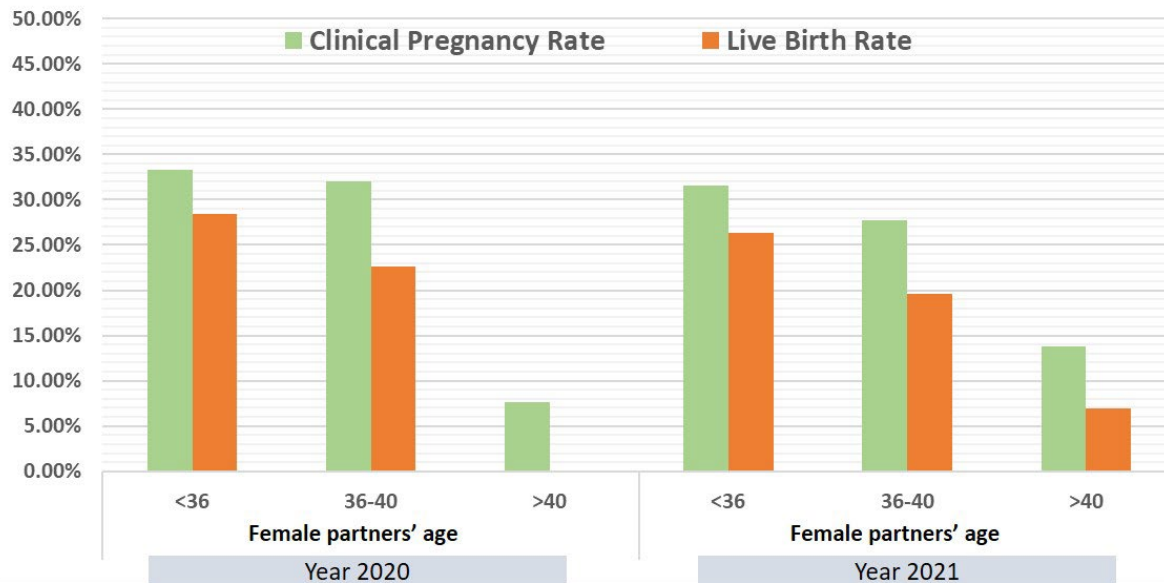
Replace 2 embryos

- ✓ ≥ 38 years at IVF or not pregnant after 2 IVF cycles And
- ✓ No livebirth before

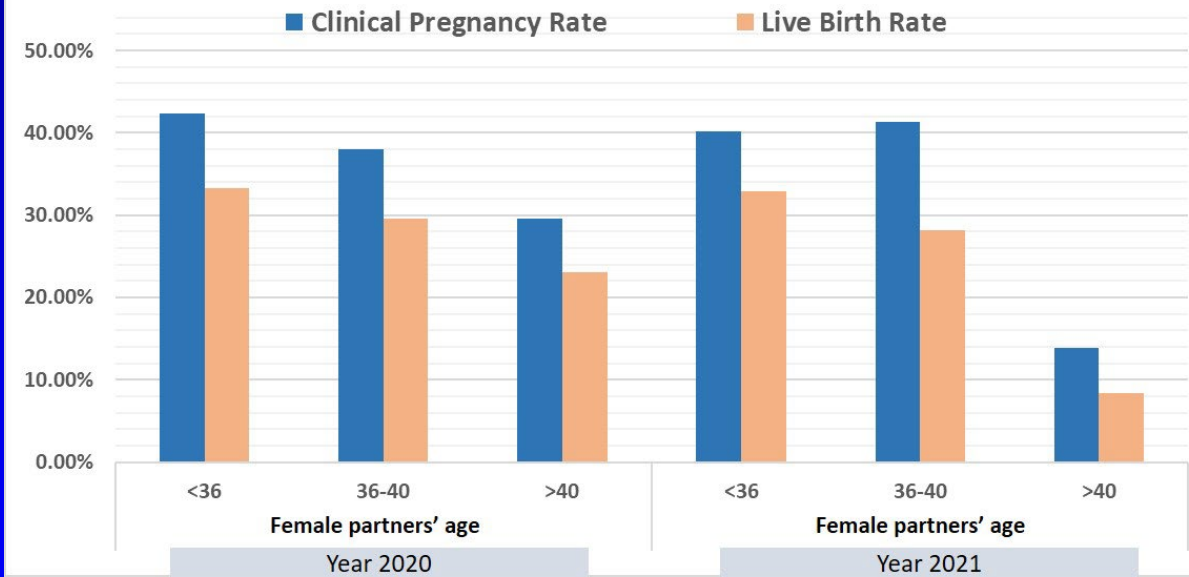
Replace **ONE** embryo
to reduce the risk of
multiple pregnancies

Success rates

Success Rate for Fresh Embryo Transfer Cycle
2020 and 2021



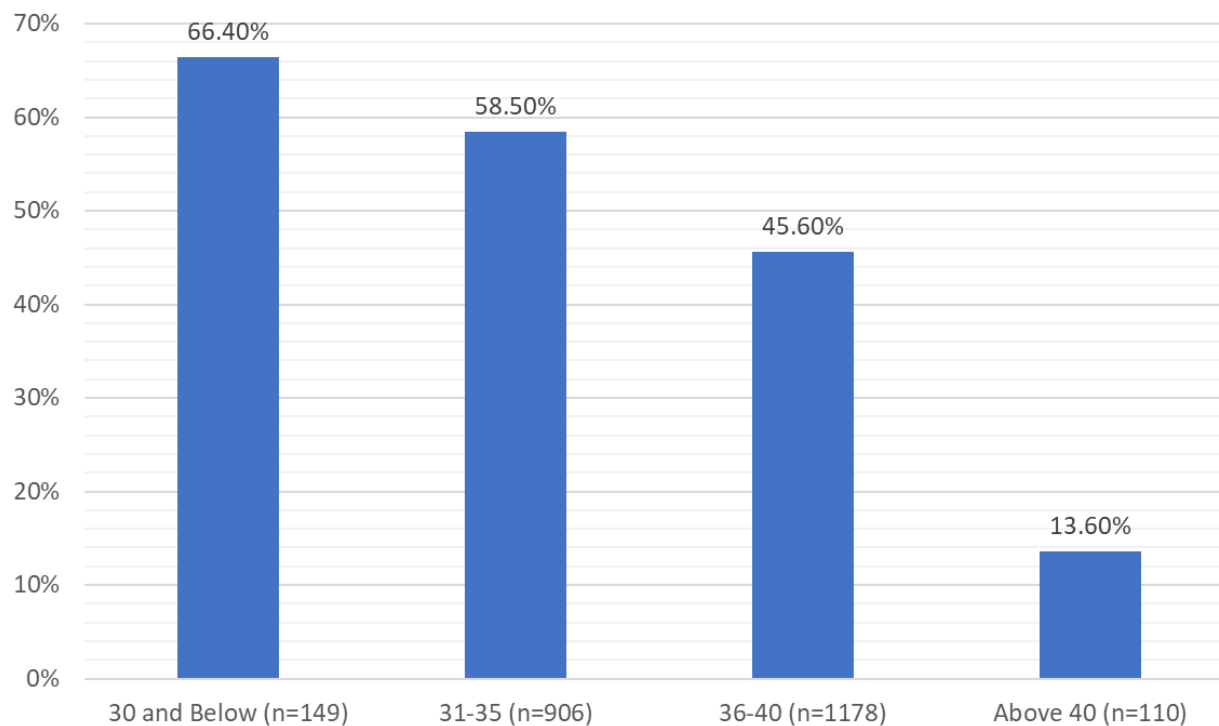
Success Rate for Frozen Embryo Transfer (FET) Cycle
2020 and 2021



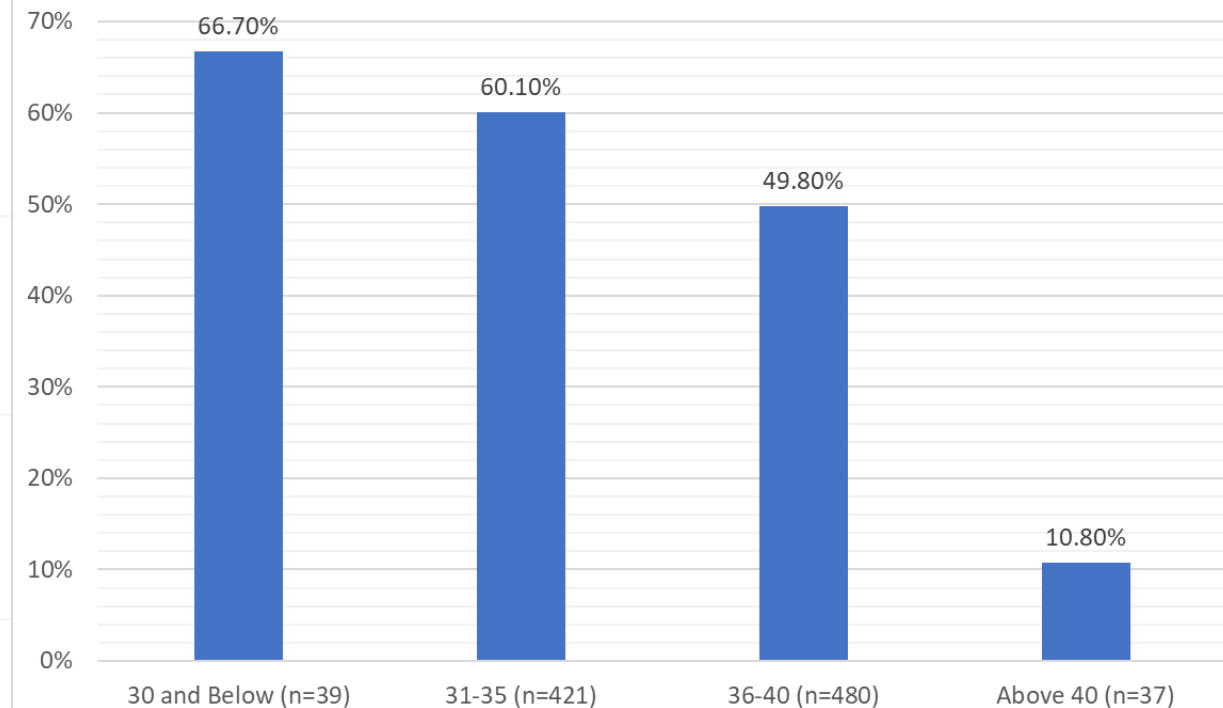
Cumulative live birth of the first IVF cycle



First IVF Cumulative Live Birth Rate, Before Policy Change (01/2013 - 08/2018)



First IVF Cumulative Live Birth Rate, After Policy Change (09/2018 - 12/2020)



Take home messages

- Age of women is an important factor. Older women should be investigated and treated early.
- Three most important investigations
 - ✓ Semen analysis
 - ✓ Serum progesterone
 - ✓ HSG and/or laparoscopy
- Effective treatment should be offered for long standing infertility



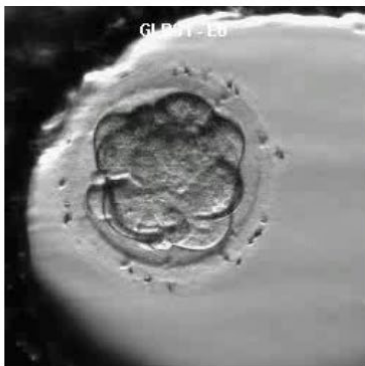
主頁

香港大學-瑪麗醫院輔助生育中心 (HKU-QMH CARE) 成立於1986年。現在，我們在廣華醫院、東區尤德夫人那打素醫院、瑪嘉烈醫院、瑪麗醫院為不同生育問題的夫婦提供生育力評估、輔導和治療。

在這些中心提供的治療都是基於最新的醫學證據。除了臨床服務外，我們亦致力於教學以及研究工作。

我們希望通過不同的社交媒體向公眾傳播有關生育健康的資訊。

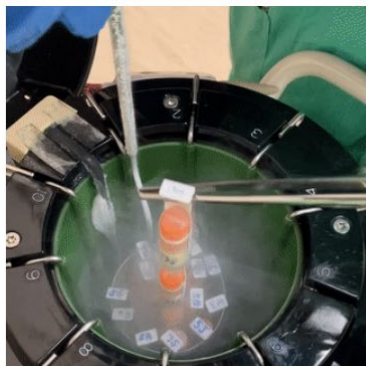
生育治療



植入前基因檢測



生育力保存



我們中心的網頁 <https://hkuivf.hku.hk/>



HKU-QMH CARE

香港大學瑪麗醫院輔助生育中心

<https://hkuivf.hku.hk>



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